

Deployment Portfolio Task 2

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Unit Code & Title: SWE40006 Software Deployment and Evolution

Semester & Submission Date: Semester 2, 24/08/2026

Declared Target Level: Distinction (2.3)

Execution Evidence:

Pass Level Task:

Setting up File Structure

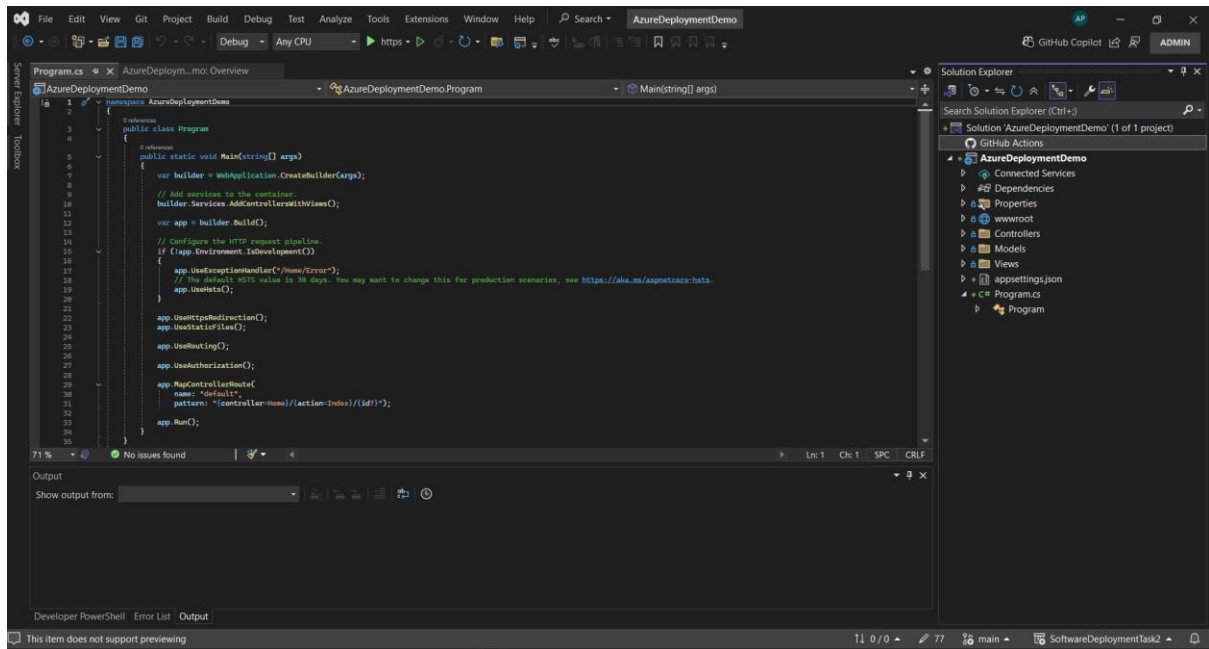


Figure 1, ASP.Net template

First, I had to download the ASP.Net which was included within the Aure development software in Visual Studio installer then created a new project using that template to create this project structure. Figure 1 also shows the Views folder which contains what the user sees and will store the html code for the Sample Application.

Azure Deployment

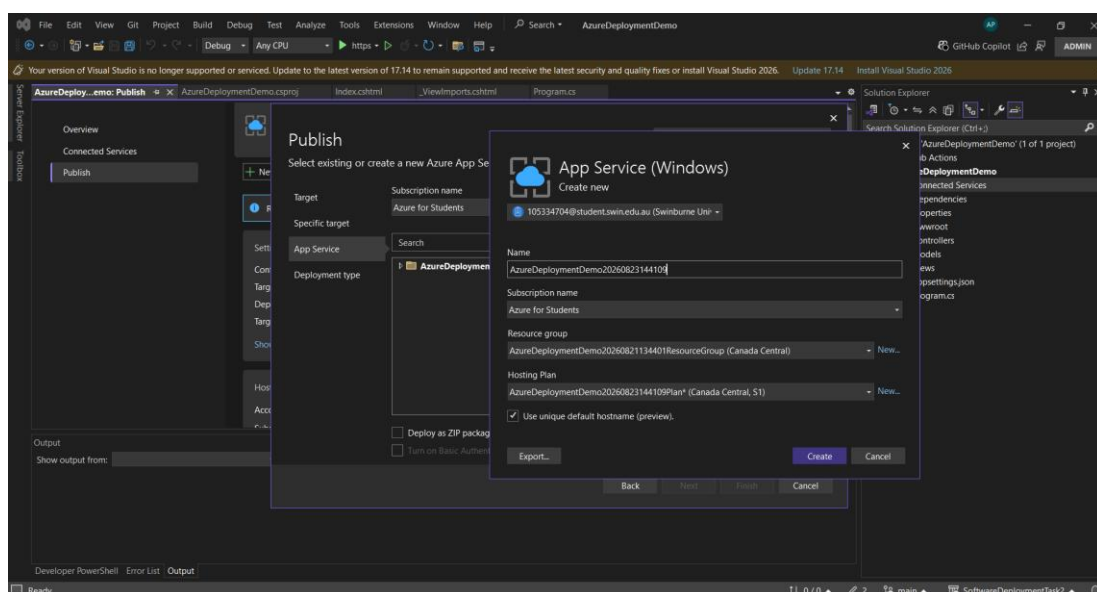


Figure 2, Azure Deployment settings

After creating the project template, I had published the project with the target as azure and to the windows system in specific target option. Lastly, a resource group was created to have a place to store the application code on the cloud, the subscription user selection had some issues with the school Azure account which required users to go onto a specific azure website for students then after verifying azure would provide a subscription which allowed users to deploy code onto the cloud.

Publishing Application

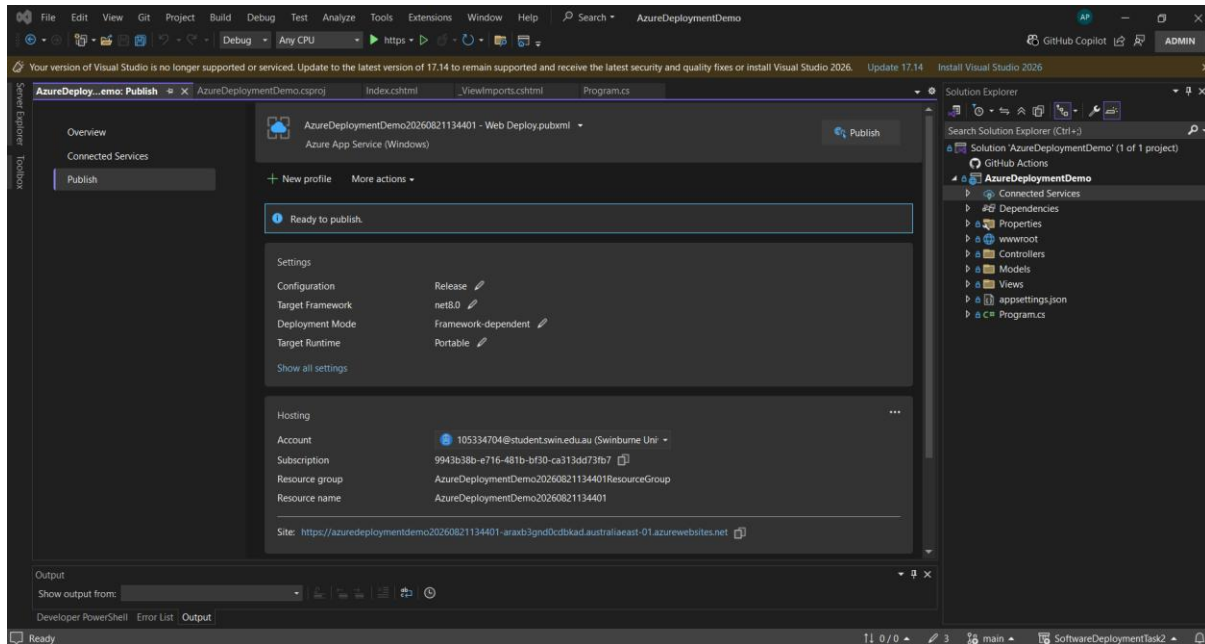


Figure 3, Publishing Application

After the resource group is created, I had navigated to the connected services menu then clicked publish on the project to publish the code onto the instance.

Application

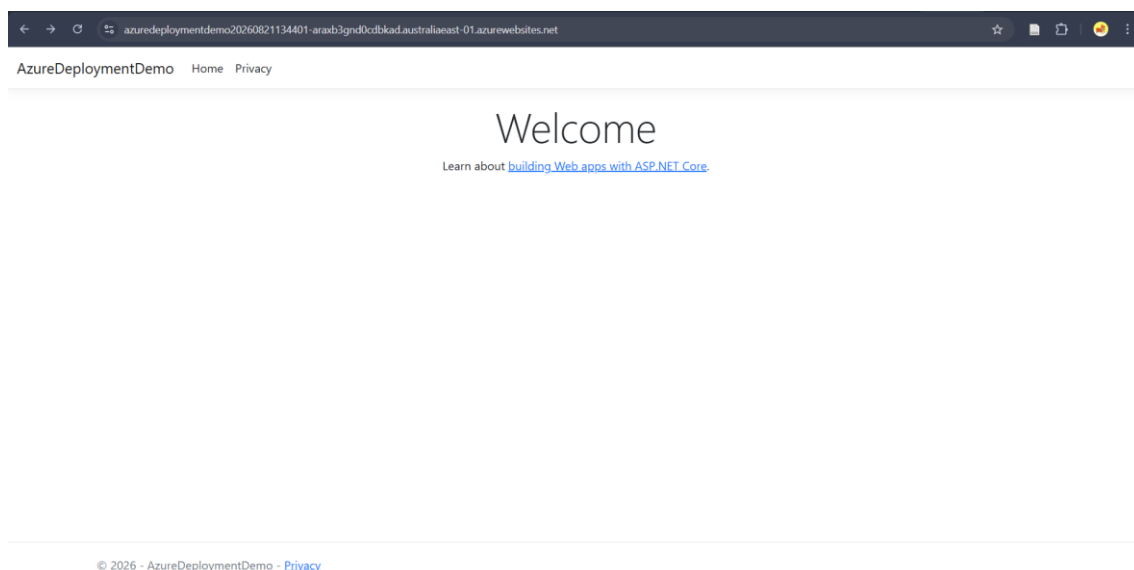


Figure 4, Application

After publishing the project, the application is available at the public site address <https://azuredeploymentdemo20260821134401-araxb3gnd0cdbkad.australiaeast-01.azurewebsites.net/> specified in the site option when publishing the code.

Credit Level Task:

Project Template

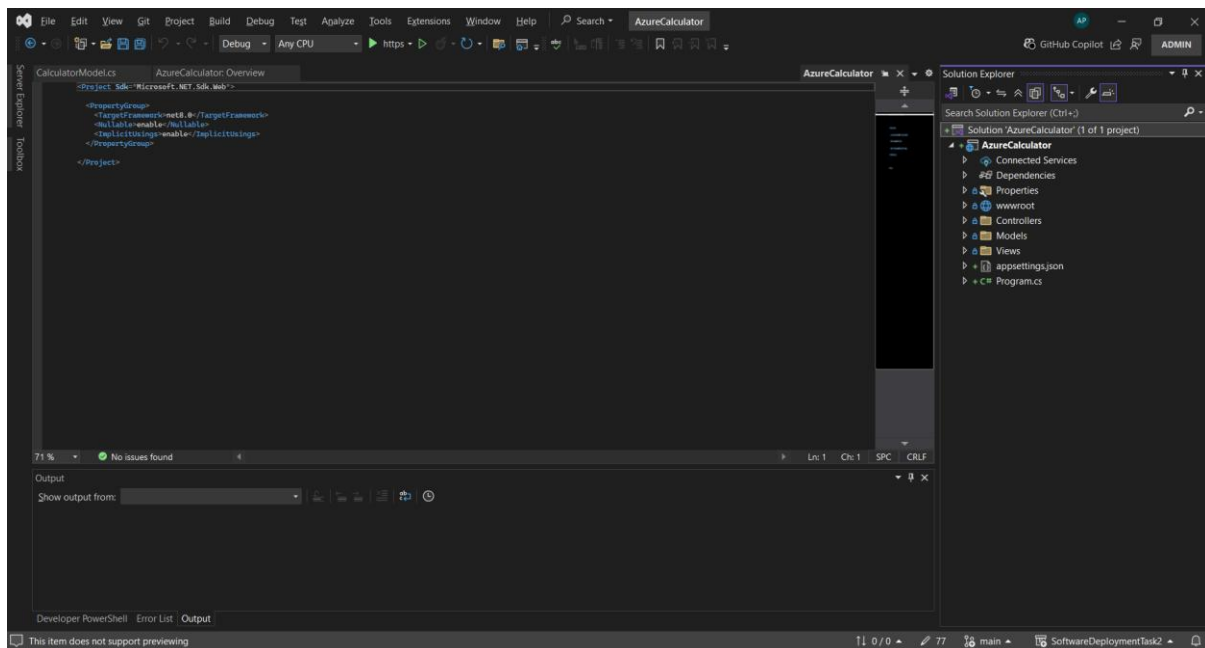


Figure 5, Calculator App

Created new project using the ASP.Net template, this new project will be used for a calculator application that will be deployed on azure.

Model Layer

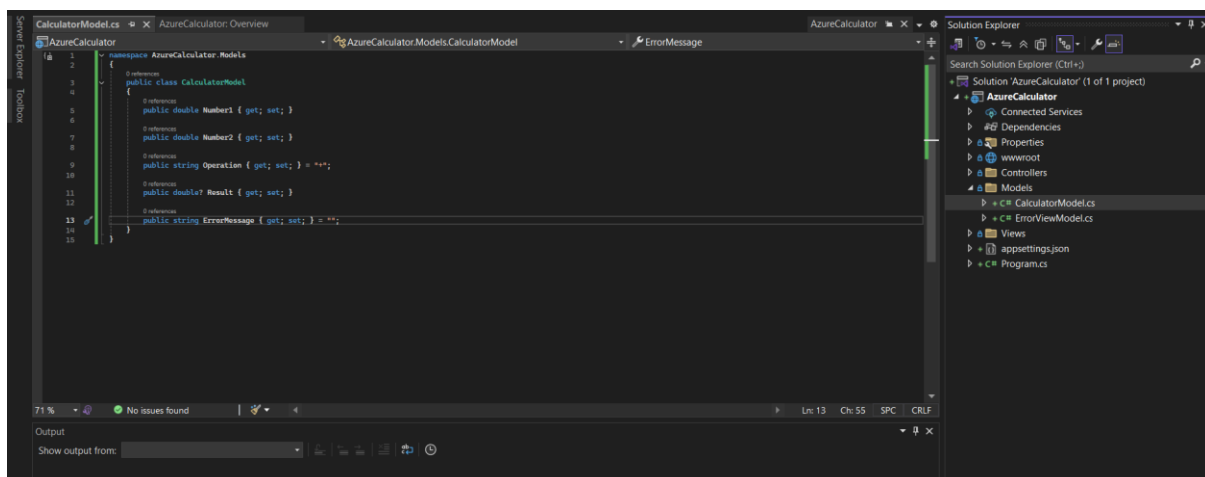


Figure 6, Model Class

Created a model class for the calculator application, the model layer of MVC will store the business data and in the case of the calculator app it will store the number and operation used.

Controller Layer

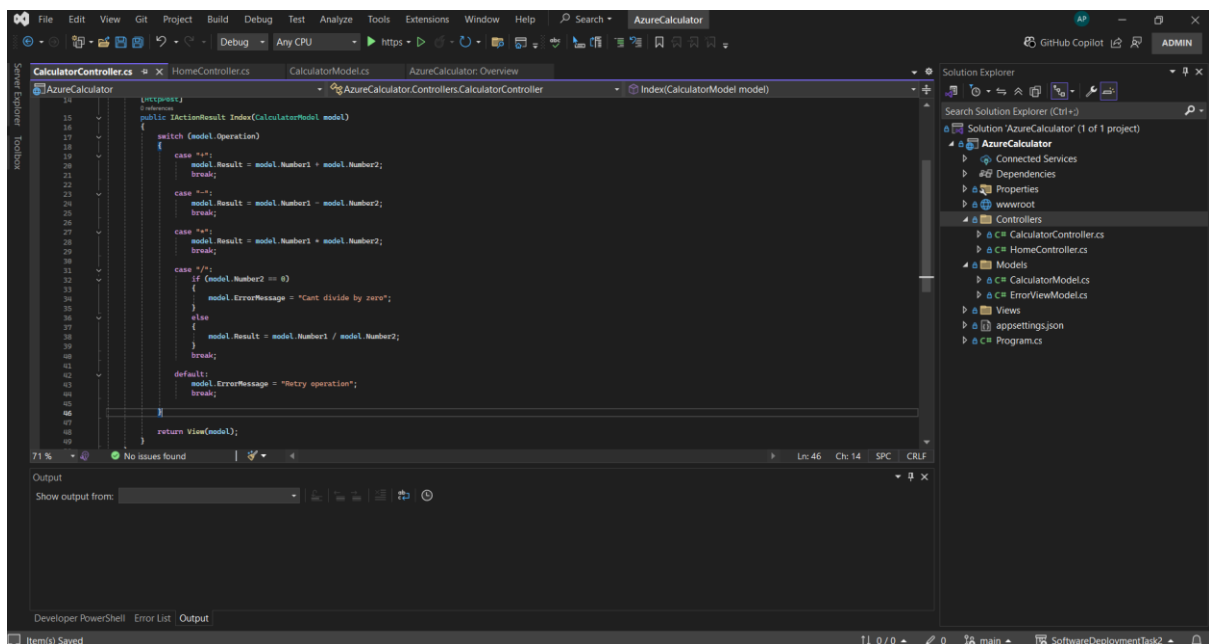


Figure 7, Model Controller

Figure 7 highlights the controller class of the model; this layer acts as the user controls where it provides the business logic and data manipulation. I developed a switch case after initialising the model object and this allows me to access the number attributes to perform the math operation on then later these numbers would be provided to the view layer.

View Layer

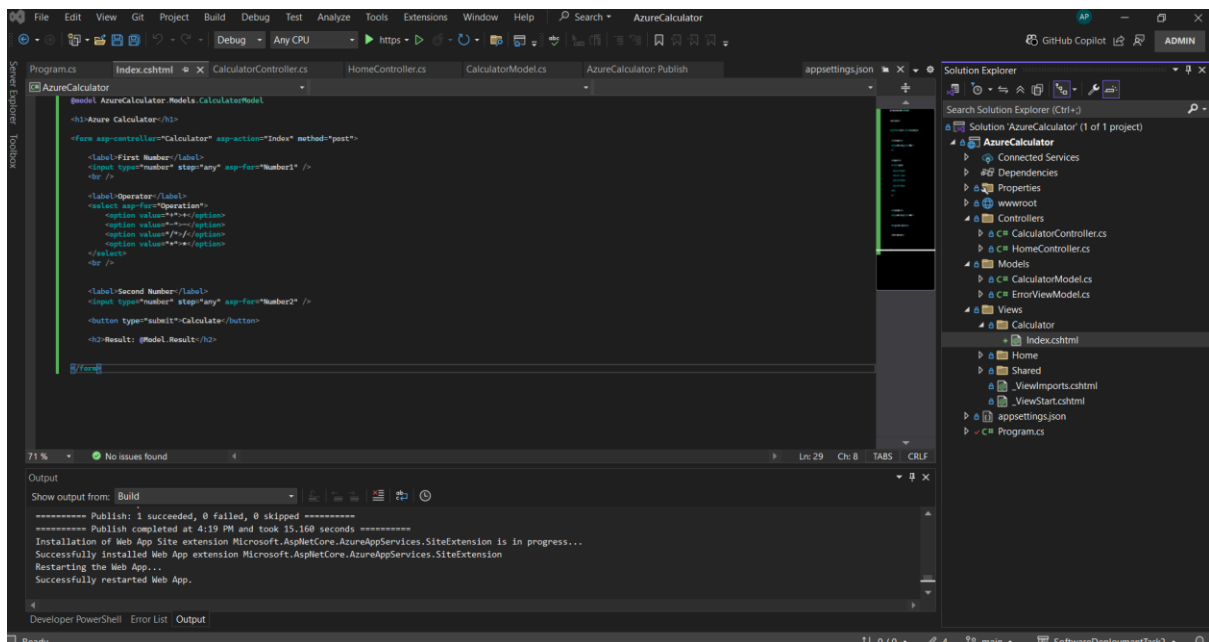


Figure 8, Calculator View

Created the front page of the calculator app by creating a folder and putting a html file which defined a form tag which uses a post request to send the values the user input into the controller layer. In each tag there is an asp line which is used to connect to ASP.NET to allow for

the sending of information. Lastly, I modified the controller route parameter in program.cs from home to Calculator to have the Calculator as the main page.

App Deployment

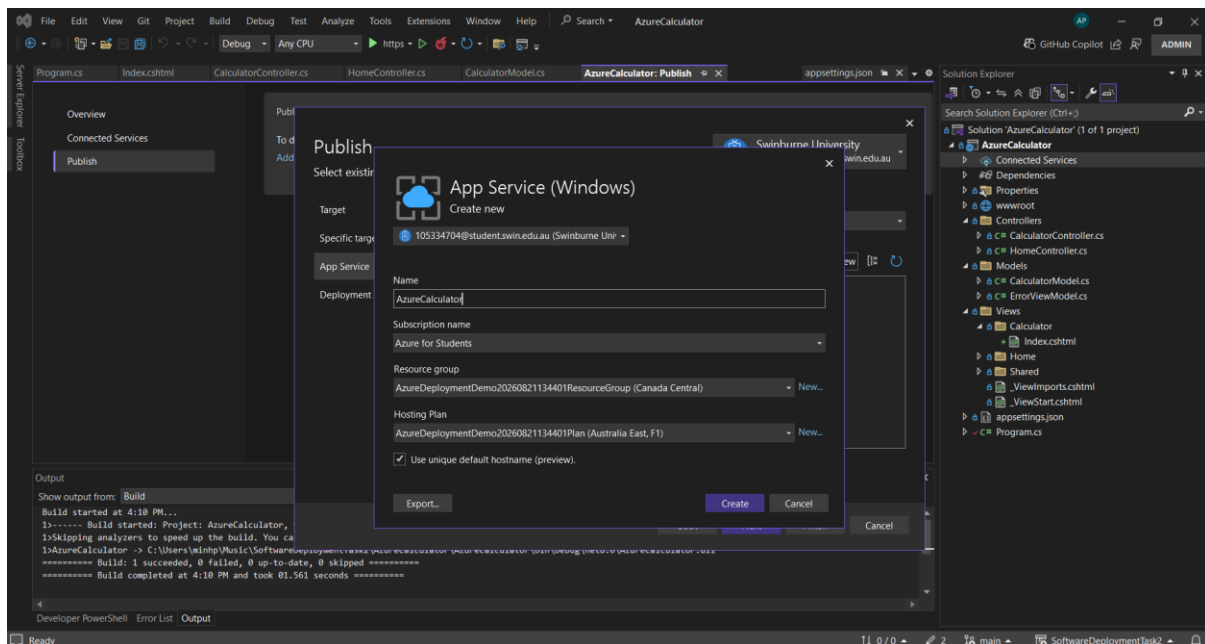


Figure 8, Deployment

Figure 8 highlights the deployment settings used to deploy the calculator application after finishing the build. I used the same resource group however I have created a new app service which is the VM used to host the application since the task is for a different grade but same assignment. Moreover, I have used the Azure as target and Azure for windows as the specific target with the hosting plan selected based on the available regions specified in the policy. Lastly, after this setup page finishes, I had clicked publish again in the publish section of Connected Services.

Live App

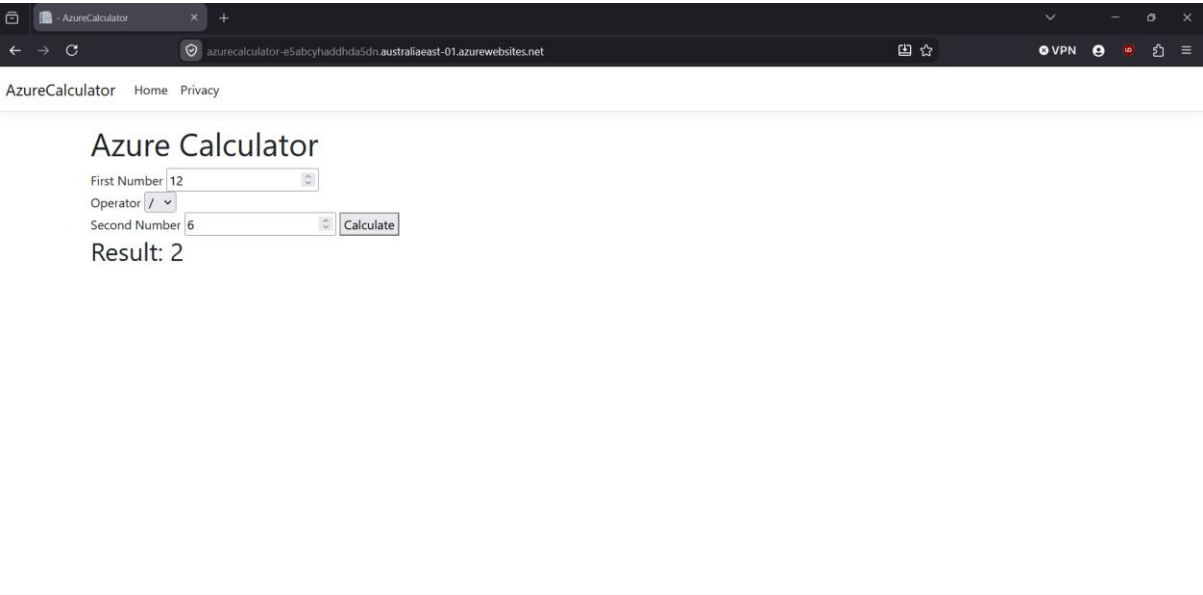


Figure 9, Working Application

Figure 9 shows the working application at this public address <https://azurecalculator-e5abcyhaddhda5dn.australiaeast-01.azurewebsites.net/>

Lifecycle & Resource Management

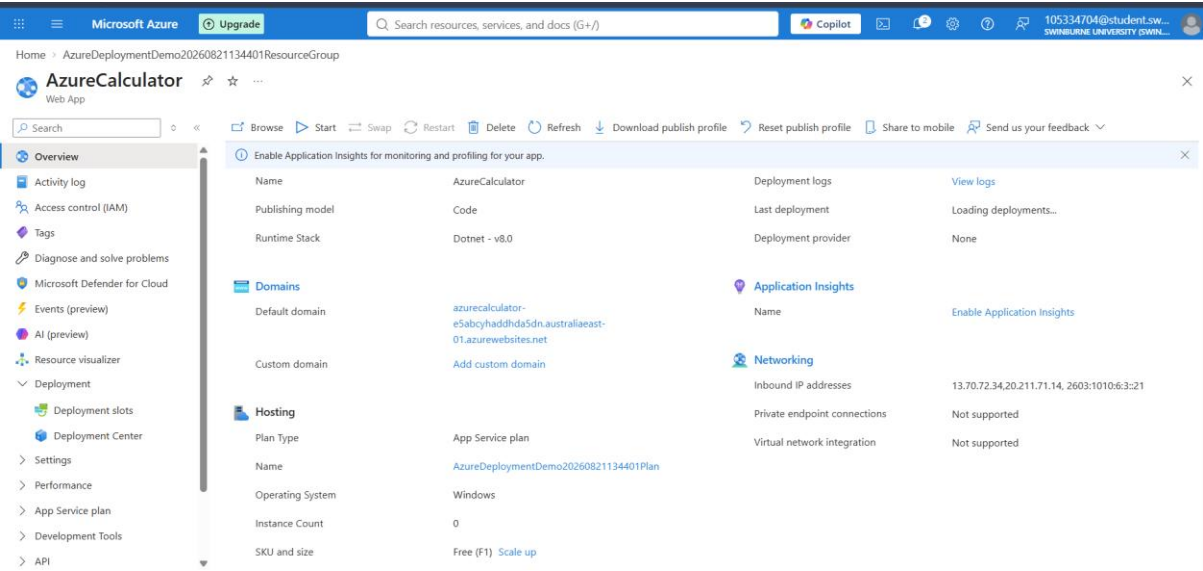


Figure 10, Lifecycle

Figure 10 highlights my efforts of responsible cloud management where the VM for the Azure cloud is stopped whenever not in use and additionally I have modified the app service plan so future VMs created will always be in the free tier which provides free limited CPU every day so the application can run without needing to be closed.

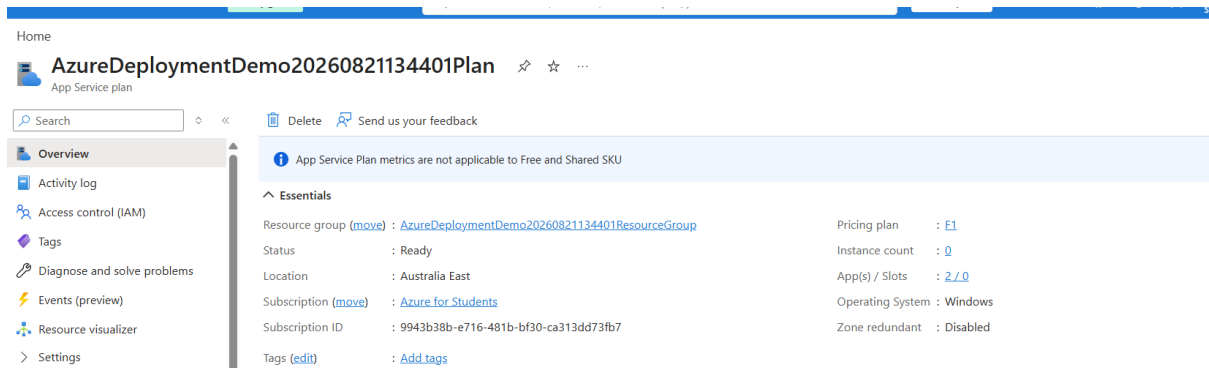


Figure 11, Free Plan

Distinction Level Task:

Environment Variable

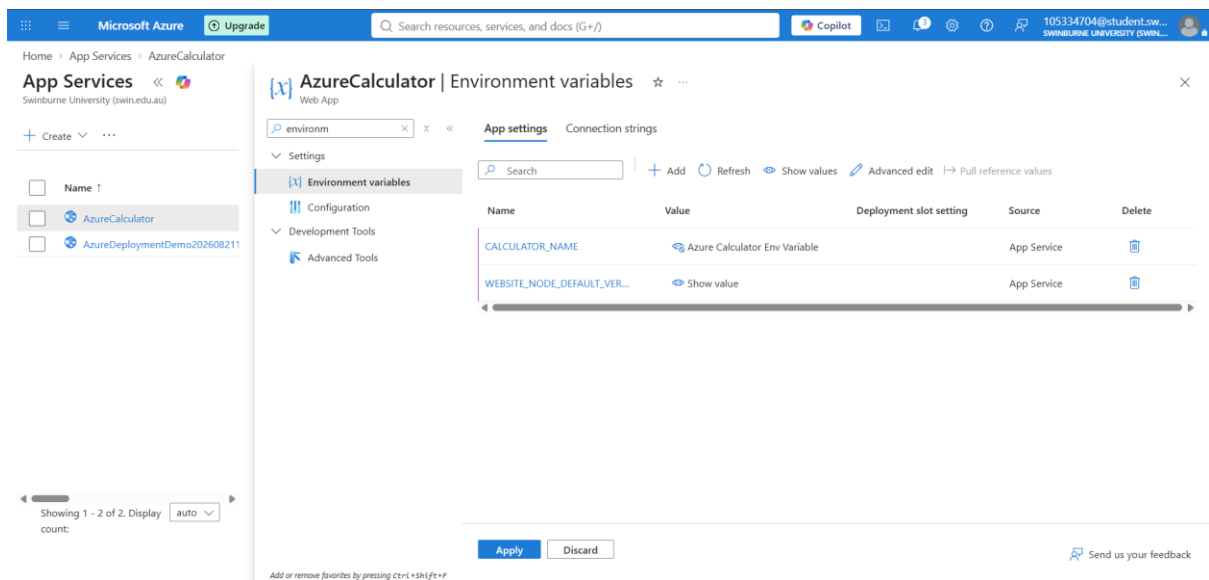


Figure 12, App Name variable

Figure 12 shows the creation of an environment variable after accessing the app service of the Azure calculator app, by setting a calculator name variable as the environment variable the application no longer needs to rely on a hardcoded name which can be modified by malicious users.

Controller Logic

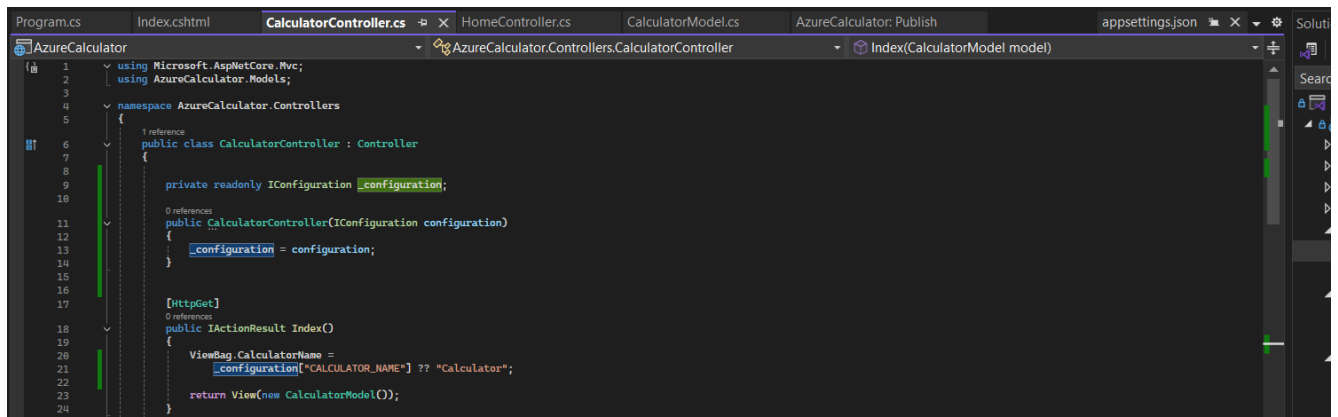


Figure 13, Reading Environment Variable

Figure 13 shows the modified controller code using ViewBag which is a way for the controller to send the data to the razor view layer, this line is put at the top and bottom because after calculation the page will be refreshed and will need to be provided with the environment variable again and if it is not provided an error will occur.

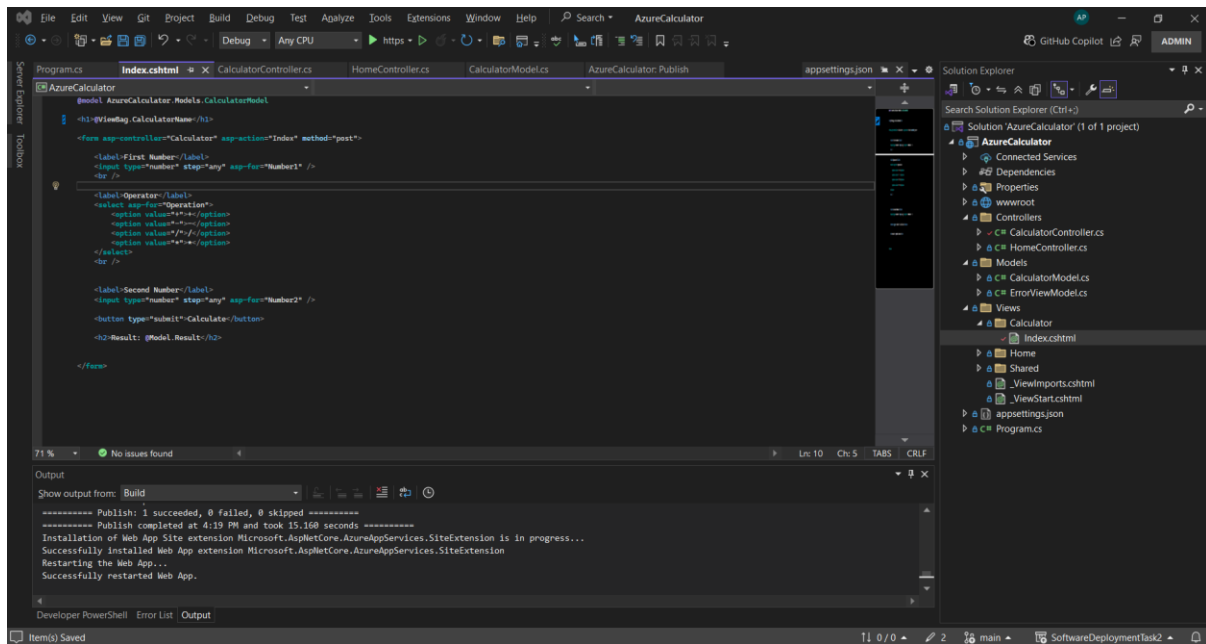


Figure 14, Modified Index

Changed the home page of calculator to not use hardcoded name but use ViewBag to get the environment variable name.

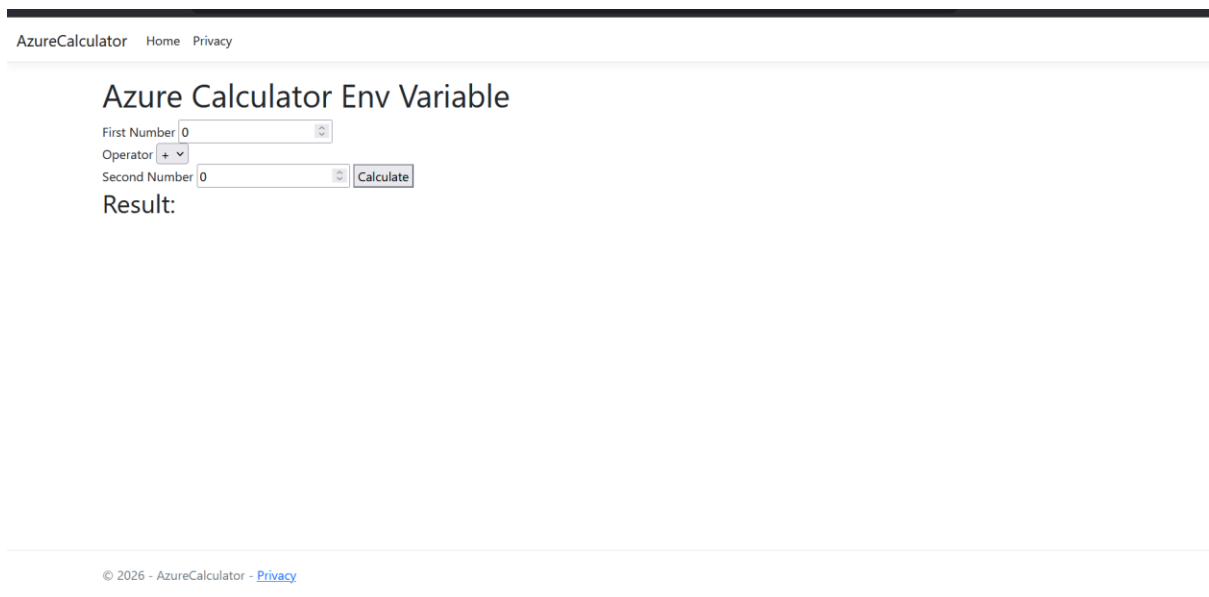


Figure 15, Result of Env

Diagnostic and Logging

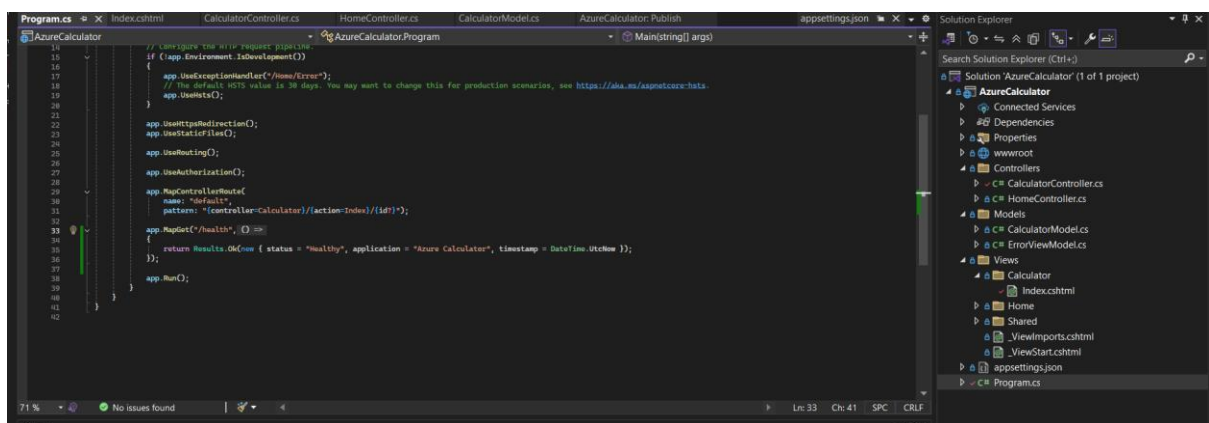


Figure 16, Health endpoint

Figure 16 highlights the health endpoint of the calculator application, whenever the user visits the homepage_url/health the application will send a HTTP 200 request to the code and if the backend code can communicate with the frontend a json response containing the application health status as well as time stamp to notify the user of the health status. This is done through the use of the mapget method which executes the health status report function whenever a get request arrives.

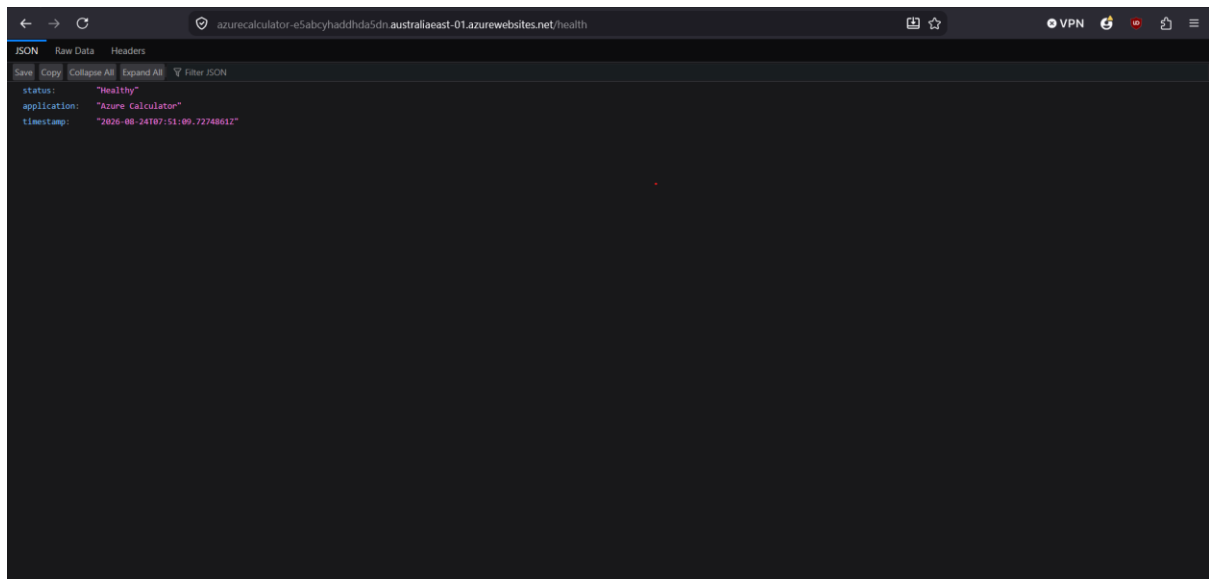


Figure 17, Healthy request

External Code & Script Repository

GitHub repository: <https://github.com/Alex-Phan-swin/SoftwareDeploymentTask2>